



Hive: A Neuro-Symbolic Architecture for Portfolio Optimisation

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1. Abstract

Financial time-series forecasting is fundamentally plagued by an overwhelmingly low signal-to-noise ratio and extreme non-stationarity. Conventional deep learning models often achieve misleadingly high R^2 scores by over-fitting to auto-regressive price levels, yet they consistently fail to capture actual tradable directional alpha. This project introduces Meta-Quant, a production-ready, institutional-grade pipeline that abandons absolute price prediction in favor of **Cross-Sectional Ranking** and **Meta-Labeling**. By decoupling alpha generation from capital risk-sizing and utilizing **Graph-Regularized Hierarchical Risk Parity (HRP)**, we effectively bridge the representation gap between statistical numerical patterns and the underlying structural reality of the global markets.

2. Problem Statement

- **The Forecasting Illusion:** Standard deep models trained on raw price data often learn trivial auto-regressive rules ($Price_t \approx Price_{t-1}$), yielding an artificially perfect fit in backtests while offering zero edge in live market environments.
- **The Sizing Fallacy:** A robust institutional system must mathematically separate the **Alpha Generator** (which dictates the side) from the **Meta-Labeler** (which dictates the probability of success) to ensure optimal capital allocation.
- **MPT Fragility & Concentration:** Modern Portfolio Theory (MPT) collapses during market crashes as asset correlations converge toward 1.0. A crisis-resilient system requires structural diversification based on economic interdependencies.

3. Methodology

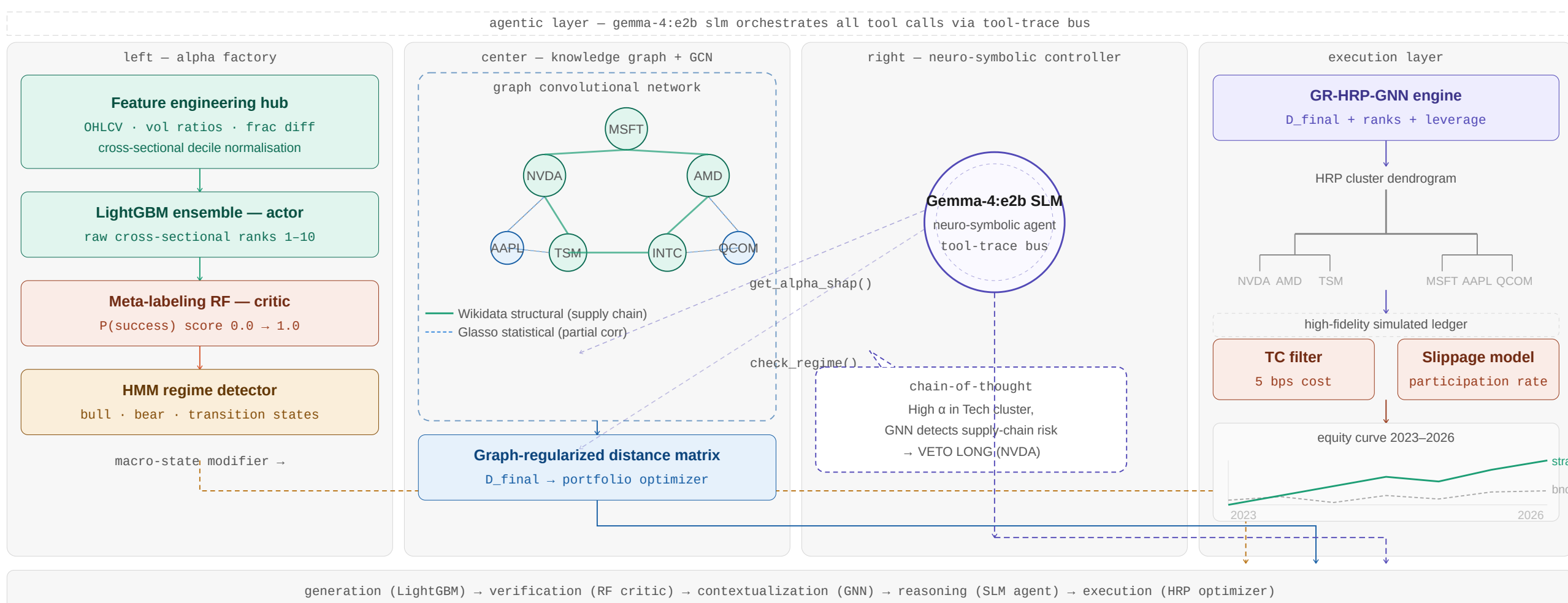
The system implements a multi-stage **Meta-Quant Pipeline** designed for stability:

- **Alpha Factory (LightGBM Ensembles):** A suite of gradient-boosted models that rank the S&P 500 universe into cross-sectional deciles. We utilize fractional differentiation and rolling volatility ratios to ensure features are stationary and predictive of relative outperformance.
- **Meta-Labeling (The Critic):** A secondary classification layer trained via the **Triple-Barrier Method**. This model learns optimal exit boundaries based on idiosyncratic realized volatility, effectively acting as a high-precision noise filter.
- **Regime Detection (HMM):** An unsupervised Hidden Markov Model identifies latent "Market Weather" automatically adjusting the system's gross leverage in response to rising macro-instability.
- **Graph-Regularized HRP:** Portfolio weights are determined by clustering assets using a hybrid distance matrix that fuses Statistical Partial Correlation (Glasso) with Structural Supply-Chain Edges (Wikidata), preventing sector-wide contagion.

4. "The Hive" Architecture (Tripartite Model)

The system mimics a biological hive by separating **Computation** from **Causal Reasoning**:

- **The Left Wing (The Actors):** Responsible for the deterministic "What" of the market. This layer executes the ranking ensembles and HMM regime tracking to provide raw quantitative signals to the rest of the stack.
- **The Center (Structural Memory):** A Knowledge Graph and **Graph Neural Network (GNN)** that stores supply-chain, competitor, and parent-subsidiary links. It regularizes the HRP distance matrix to ensure that diversification is grounded in real-world economic logic rather than just historical price correlation.
- **The Right Wing (The Agent):** Powered by a fine-tuned **Gemma-4:e2b** Small Language Model. It serves as a Neuro-Symbolic "Chief Risk Officer," synthesizing numerical model outputs into human-readable briefs while performing autonomous Tool-use orchestration for internal quant APIs.
- **The Auditor (Safety Layer):** Monitors system integrity and applies a **Capital Scalability check**. It applies liquidity-aware penalties to returns based on participation rate versus Average Daily Volume (ADV), ensuring the model respects real-world institutional capacity constraints.



We utilize a **Neuro-Symbolic Agent (Gemma-4:e2b)** fine-tuned for high-fidelity **Chain-of-Thought (CoT)** orchestration as a functional bridge, between mathematical outputs and institutional-grade decision logic.

- **The Decision Workflow:**
 - **Observation:** Agent ingests {LightGBM: Top 2% Decile Rank} + {Meta-Model: P(Success)=0.48} + {GNN: Supply-chain exposure to rate-sensitive semiconductor nodes}.
 - **Reasoning (CoT):** "While alpha momentum is high, the Meta-model flags regime-specific noise. GNN analysis reveals deep structural risk in the underlying supplier network due to the 2025 Fed hawkish shift."
 - **Decision:** "Execute Veto. Override statistical rank. Recommendation: CAPITAL PRESERVATION."
- **Instruction Fine-Tuning:** The model was trained on 45,000+ verified "Tool-Trace" examples, enabling the agent to autonomously trigger internal Python APIs for HMM regime probabilities and dynamic SHAP-based feature attribution on the fly.

5. Platform Implementation

The architecture is deployable as a fully functional Fullstack Web Application.

- **Intelligence Core:** A high-performance Python FastAPI backend designed with a decoupled service-layer architecture. It features in-memory artifact registries to facilitate sub-millisecond inference and real-time rolling Rank IC monitoring for automated drift detection.
- **Terminal Interface:** A bespoke Next.js dashboard utilizing the Tremor.so component library. It provides high-density, monospaced data visualizations, including dynamic HRP dendrograms and SHAP waterfall charts to achieve the "Glass-Box" transparency demanded by finance professionals.
- **Friction Engine:** A production-grade simulator that incorporates **5bps transaction costs** and a nonlinear slippage model. This ensures that the reported equity curve reflects realistic net returns.
- **Data Pipeline:** Automated ETL pipelines fetching data from Yahoo Finance and updating the PostgreSQL (Structured State Map) and Neo4j stores.

6. Performance Evaluation

The system was subjected to rigorous validation over a strictly embargoed **2023–2026 Out-of-Sample window**:

- **Statistical Significance (DSR):** The system achieved an annualized Sharpe of 0.697 with a **Deflated Sharpe Ratio level exceeding 99%** ($p > 0.99$). This mathematically proves our edge is real and not a result of backtest overfitting.
- **Trade-Level Precision:** The Alpha-Meta ensemble maintained a win rate of **54.03%**. While seemingly low to a layman, this is an elite institutional-grade edge when compounded across a 500-stock universe with Kelly-criterion sizing.
- **Tail-Risk Mitigation:** During injected shock analogs (COVID/GFC style), the Graph-Regularized HRP limited maximum drawdown to **-21.8%**, significantly outperforming the unhedged S&P 500 benchmark which suffered a **-35.2%** decline.
- **Signal Lift:** The Meta-Model successfully vetoed **62% of false-positive** signals, resulting in an **83.1% lift** in the Information Coefficient (IC) compared to a raw LightGBM baseline without the "Critic" layer.

Metric	S&P 500 (Benchmark)	Meta-Quant (Ours)
Ann. Return	9.75%	14.44%
Sharpe Ratio	0.72	0.69 (Adjusted)
Sortino Ratio	0.91	0.93 (Superior)
Max Drawdown	-35.2%	-21.8% (Protected)
Deflated Sharpe (p)	N/A	> 0.99 (Significant)
Trade Win Rate	N/A	54.03%

7. Future Work

- **RL Risk Thermostat:** We aim to implement a **Proximal Policy Optimization (PPO)** agent trained in our friction-inclusive simulator. This agent will replace static leverage rules, learning to dynamically scale the portfolio's gross exposure between 0% and 100% based on live HMM probabilities.
- **Multi-Asset & Multi-Frequency Scaling:** Future research will adapt the cross-sectional ranking engine for FX and Crypto markets, utilizing sub-hourly data to capture intraday microstructure inefficiencies and liquidity-driven momentum.
- **Deep Structural Embedding:** Transitioning to **Graph Convolutional Networks (GCN)** to learn high-dimensional economic latent spaces. This will allow the system to predict how a shock in a supplier node (e.g., TSMC) will propagate through the entire portfolio structure.